

Emre Kerman

Computer Engineer

İzmir, Turkey

+90 541 264 0344

emrekerman02@gmail.com

github.com/emrekerman

linkedin.com/in/kermanemre

SUMMARY

Computer Engineer from İzmir and recent Dokuz Eylül University graduate with experience in software development, machine learning, embedded systems, and full-stack web development. Experienced in building IoT and computer vision systems, developing frontend and backend applications, and working with Linux and open-source technologies.

EDUCATION

Dokuz Eylül University

Bachelor of Science in Computer Engineering

Izmir, Turkey

2021 – 2025

Yunus Emre Anadolu Lisesi

High School Diploma

Izmir, Turkey

2018 – 2021

EXPERIENCE

Software Engineering Intern

June 2025 – Sept. 2025

İndas Teknoloji Ticaret AŞ.

Izmir, Turkey

- Engineered a smart parking platform using ESP32-CAM nodes and OCR (PaddleOCR/Tesseract) for automated vehicle entry tracking, billing, and balance management via Next.js REST APIs.
- Built an edge-first PPE and human detection system on ESP32-CAM using quantized FOMO models, C++/Arduino, and Edge Impulse, achieving 120–250ms inference latency, 18 FPS live streaming, and 5–6 FPS on-device inference through dual-buffering and PSRAM optimization.
- Developed web frontends using Next.js and React for real-time bounding box overlays, detection logs, and balance dashboards.

Software Engineering Intern

Nov. 2025 – Dec. 2025

Dokuz Eylül University Computer Engineering Dept.

Izmir, Turkey

- Developed "Restock", a self-checkout platform integrating physical store kiosks with mobile devices for real-time session handover.
- Implemented atomic DB transactions using Drizzle ORM and SQLite alongside custom React hooks for barcode scanning.
- Configured role-based auth (JWT) via Next.js Middleware and built administrative monitoring dashboards.

PROJECTS

Early Frost Detection | Go, Python, C, TensorFlow, Keras, NumPy, ESP-32

2024 – 2025

- Developed a machine learning-based IoT solution to predict ice formation on refrigerator cooling fins for an Arçelik use case.
- Designed and assembled a hardware refrigerator prototype, integrating ESP32 microcontrollers and DHT22 sensors to collect environmental data over Wi-Fi.
- Performed data preprocessing, feature engineering, and machine learning model training on the collected data.
- Built a real-time web server and front-end monitoring platform using Go.

Hand Gesture Detection | Python, TensorFlow, Keras, NumPy, OpenCV

2024

- Constructed two computer vision models (a lightweight CNN and a ResNet-like architecture) to classify gestures via webcam.
- Curated and preprocessed a custom hand-gesture dataset for model training.
- Implemented an efficient training pipeline utilizing Model Checkpoints to prevent data loss.
- Integrated live webcam input with UX enhancements for real-time inference.

Game Platform & Storefront | C#, .NET Core, PostgreSQL

2023

- Designed and implemented the complete database architecture for a digital game distribution platform.
- Developed backend application logic to support user authentication, game publishing, and purchasing.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C/C++, Go, C#, SQL, Rust

Web: Next.js, React, React Native, Astro, Tailwind CSS, REST APIs

Backend & Databases: PostgreSQL, SQLite, Drizzle ORM, .NET Core, OAuth/JWT, Swagger/OpenAPI

AI, Vision & Edge: TensorFlow, Keras, TFLite, OpenCV, Edge Impulse, ESP32/ESP32-CAM, PaddleOCR

Systems & DevOps: Linux, Docker, Kubernetes, Git, Nix/NixOS

Spoken Languages: Turkish (Native), English (C1), German (A1)